

EXPERIENCE

- **University of Melbourne** Melbourne, Australia (Remote)
Machine Learning Engineer (Contract) Dec 2023
 - **Multimodal Pipeline:** Built an end-to-end **multimodal Twitch analytics pipeline** processing **13+ TB** of video, audio, and live chat across **40** streamers to produce minute-level **sentiment, toxicity, and emotion** signals.
 - **Computer Vision:** Implemented **GPU-accelerated face identity filtering** using **InsightFace** embeddings and cosine similarity to robustly isolate streamer face frames across VOD segments; classified emotions per-chunk with **DeepFace**.
 - **NLP & Audio Analysis:** Integrated **faster-whisper** for speech transcription and engineered audio features (pitch, RMS loudness, bitrate). Built a multi-model scoring layer combining **GoEmotions**, **Detoxify**, and **HateSonar** for chat and transcript analysis, achieving **~92%** accuracy on emotion classification.
 - **Tooling & Infrastructure:** Productionized the system with a **CUDA-enabled Docker** environment; built a **Flask-based human-in-the-loop face embedding builder** to bootstrap per-streamer identity recognition for the CV pipeline.
- **Facts-n-Data** Remote
Software Development Intern Mar 2024 – Apr 2025
 - **E-Commerce Automation:** Built and maintained **30+** data scraping and automation scripts across **Amazon**, **FirstCry**, and **Myntra**, collecting data on **10,000+** products to enable large-scale analysis.
 - **Analytics & Monitoring:** Developed scripts for **BSR tracking**, **search term analysis**, and **real-time stock monitoring** to drive data-informed business decisions.
 - **Internal Tooling:** Refactored legacy codebases for improved efficiency and built internal tools including **user authentication** systems and **employee management dashboards**.
- **BLITZ, The Computer Science Society** Keshav Mahavidyalaya, Delhi University
Secretary Aug 2025 – Present
 - **Web Development:** Designed and developed the **official society website** for both annual cycles, reaching **5,000+** students and serving as the primary digital presence for events and announcements.
 - **Volunteer Management:** During flagship events, **managed all volunteers** on-ground, assigning tasks and ensuring smooth execution across all event verticals.
- **PM Shri Kendriya Vidhyalaya** Vigyan Vihar, Delhi
Publication Captain Apr 2023 – Mar 2024
 - **School Blog:** Developed and maintained the **entire school blog**, serving as the primary platform for school news and student content.

PROJECTS

- **Custom ESP32 Drone – Full-Stack Hardware Build** Personal Project
 - **Flight Firmware:** Flashed and configured **ESP-FC** flight-controller firmware on **ESP32**; reverse-engineered the FlySky FS-CT6B transmitter's USB protocol to read RC channel data directly.
 - **Signal Pipeline:** Converted **PWM to PPM** via an Arduino Nano intermediary, then migrated to **RMT-based** RC channel reading on the ESP32 for lower-latency control input.
 - **Flight Stability:** Implemented a **complementary filter** for sensor fusion, fixed yaw-mixing sign errors, and debugged ISR race conditions and Wi-Fi radio interference affecting in-flight stability.
 - **Live Tuning:** Enabled real-time **PID gain tuning** via AUX-channel knobs on the transmitter for on-the-fly flight characteristic adjustments.
- **INT8 Quantized Inference Engine – On-Device ML in C** [Blog](#) | [GitHub](#)
 - **From-Scratch Training Pipeline:** Implemented a full **MLP training pipeline** (forward/backward pass, **Adam optimizer**) in pure **NumPy** with zero ML-framework dependencies, training a **668-parameter**, **30-16-8-4** network to **97%** test accuracy on IMU-derived gesture features.

- **INT8 Quantization:** Designed a **symmetric per-tensor INT8** quantization scheme with **INT32** bias accumulators and data-calibrated inter-layer requantization, compressing the model to a **1.1 KB** binary with under **2%** accuracy loss.
- **Header-Only C Runtime:** Built a dependency-free, **header-only C inference engine** (~250 LOC, no TFLite Micro/Edge Impulse) that loads the quantized binary from **SPIFFS** and classifies gestures from live **MPU6050** IMU data in real time on an **ESP32**; model-agnostic for any dense-only MLP, and being extended toward **sensorless torque estimation** for robotic actuators.
- **Planq** – Exam Preparation Platform [Website](#) | [GitHub](#)
 - **Question Bank:** Curated a database of **47,000+** previous year questions spanning **6 All-India entrance exams**, serving **240+** registered students with **500+** concurrent live sessions at peak load.
 - **Smart Scheduling:** Implemented an **Anki-like spaced repetition** algorithm that schedules topics based on individual student performance and preparation analytics.
- **Donut Shop** – Home Server Infrastructure
 - **Cluster Setup:** Built a **2-node Linux cluster** in a **master-worker** configuration orchestrated with **k3s** and **Longhorn** for persistent storage.
 - **Services:** Deployed and managed **NAS**, media streaming (**Plex/Jellyfin**), **VPN**, and various containerized services.

SKILLS

- **Languages:** Python, C, C++, JavaScript, SQL
- **Embedded & Hardware:** ESP32 / Arduino, FreeRTOS, RC & flight-control firmware, PWM/PPM signal processing, sensor fusion, motor control (BLDC, SimpleFOC, PID tuning)
- **Machine Learning:** INT8 quantization, on-device inference, faster-whisper, InsightFace, DeepFace
- **Databases:** MongoDB, Supabase, SQLite
- **Frameworks:** Flask, Next.js, Tailwind CSS, Selenium, Pygame, DearPyGui
- **DevOps:** Docker, k3s, Linux, Git

EDUCATION

- **Keshav Mahavidyalaya, Delhi University** Pitampura, Delhi
Bachelor of Science in Computer Science; CGPA: 6.88 (Till 3rd Semester) *Aug 2024 – May 2028*
- **PM Shri Kendriya Vidhyalaya** Vigyan Vihar, Delhi
10+2, PCM; 88%/73% *April 2012 – March 2024*

ACHIEVEMENTS

- **Pariksha Pe Charcha 2024:** Presented **Anisha**, an AI-powered therapist with real-time emotion detection, directly to the Hon'ble Prime Minister of India.
- **Brainwave 2.0 Hackathon (2026):** Secured **3rd place in the Hardware track** and **8th overall** as Finalists at the 36-hour hackathon by Dept. of IT, DTU, in association with AIMS-DTU.
- **Smart India Hackathon 2025:** Secured **1st Runner-Up** in the Internal Round at Keshav Mahavidyalaya, Delhi University.